

CS1200 / CSCI E-120 Lecture 6

Randomized algorithms

Fall 2026

Recommended reading: Hesterberg–Vadhan, Section 10.1-10.4

Announcements

- Mid-term next Wed (Lec 1-5, Pset 0-2, Section 0-2).
- Pset 2 due tomorrow.
- Pset 3 released.

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- Clarification on in-class reflection: we only drop grade if we find the combination (*did well on concept X in pset, did bad on concept X in in-class reflection*).
 - For example, if you did bad on pset 1 in class reflection, and also did not correctly discuss stability in pset 1, your grade will not be affected.

Course so far

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- Today: randomization.

Median of key–value pairs

Let $A = ((K_0, V_0), \dots, (K_{n-1}, V_{n-1}))$, with $n \geq 1$. A median is a key-value pair (K, V) such that at most $\lfloor n/2 \rfloor$ keys are $< K$ and at most $\lfloor n/2 \rfloor$ keys strictly are $> K$.

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Q: Median of $((1, a), (3, b), (2, c), (4, d))$?

$(3, b)$ **and** $(2, c)$.

Why median?

Consider the keys

2, 3, 5, 4, 5, 3, 6.

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Q: What happens if the last key changes from 6 to 600?

mean = $\frac{622}{7} \approx 88.9$, median = 4.

Medians are less sensitive to extreme values than means.

The selection problem

Input: a nonempty array A of n key–value pairs with keys in $\mathbb{N}$, and a rank $i \in [n] = \{0, \dots, n - 1\}$.

Output: a key-value pair whose key is an $(i + 1)$ st-smallest key.

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$$|\{k : K_k < K_j\}| \leq i, \quad |\{k : K_k > K_j\}| \leq n - i - 1.$$

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Q: Which rank means a minimum? A maximum? A median?

$$0, \quad n-1, \quad \left\lfloor \frac{n-1}{2} \right\rfloor.$$

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Theorem

There is a **randomized algorithm** called *QuickSelect* that always solves SELECTION correctly and has worst-case *expected* running time $O(n)$.

Random choices

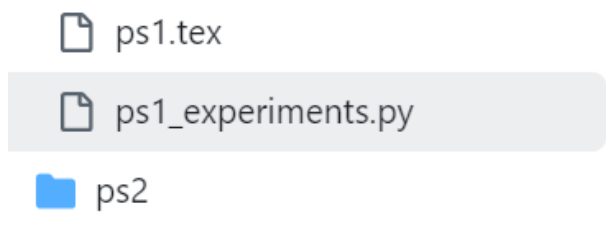
ps1.tex

ps1_experiments.py

ps2

```
25 def gen_random_arr(n, U):  
26     """generates random array given n and universe U"""  
27     arr = [(random.randint(0, U - 1), "foo") for _ in range(n)]  
28     return (arr, U, n)
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```

Python's `random.randint(0, U-1)` draws an integer from $[U] = \{0, \dots, U - 1\}$.

Randomness in pseudocode

Add the command in the pseudocodes we have seen so far:

$$u = \text{random}(U), \quad U \geq 1.$$

Q: What does this command do?

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- It assigns u a uniformly random element of $[U]$.
- Every value has probability $1/U$.
- Different calls produce independent random values.

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- Every value has probability $1/U$.
- Different calls produce independent random values.

We count this command as one basic operation.

* We have not introduced precise model of computation yet; we'll do so after midterm 1.

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Choose a uniformly random key-value pair, and call its key as *pivot*.

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- All keys in $A_{<}$ are less than all keys in $A_{=}$.
- All keys in $A_{=}$ are less than all keys in $A_{>}$.

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- All keys in $A_{<}$ are less than all keys in $A_{=}$.
- All keys in $A_{=}$ are less than all keys in $A_{>}$.

Calculating the sizes of these arrays tells you where to look for rank i key.

QuickSelect: choosing and partitioning

Algorithm 1: QuickSelect(A, i)

Input: $n = |A| \geq 1$, keys in $\mathbb{N}$, and $0 \leq i < n$.

Output: A rank i key-value pair rank.

- 1 if $n = 1$ then return $A[0]$
 - 2 Set $p = \text{random}(n)$ and pivot $= K_p$
 - 3 Set $A_{<} =$ _____
 - 4 Set $A_{=} =$ _____
 - 5 Set $A_{>} =$ _____
 - 6 Set $n_{<} = |A_{<}|$, $n_{=} = |A_{=}|$, and $n_{>} = |A_{>}|$
-

Q: Which key-value pairs belong in $A_{<}$?

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- 1 if $n = 1$ then return $A[0]$
 - 2 Set $p = \text{random}(n)$ and $\text{pivot} = K_p$
 - 3 Set $A_{<} =$ the key-value pairs of A with keys $<$ pivot
 - 4 Set $A_{=} =$ _____
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Q: How should we handle key-value pairs tied with the pivot?

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Q: What remains for $A_{>}$?

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 - 6 Set $n_{<} = |A_{<}|$, $n_{=} = |A_{=}|$, and $n_{>} = |A_{>}|$
-

The three sub-arrays include all the key-value pairs, and $n_{<} + n_{=} + n_{>} = n$.

QuickSelect: choosing the recursive call

View of the sub-arrays in the sorted order:

$\underbrace{A}_{A_{<}} \quad \underbrace{A}_{A_{=}} \quad \underbrace{A}_{A_{>}} \quad .$
 $n_{<} \text{ key-value pairs} \quad n_{=} \text{ key-value pairs} \quad n_{>} \text{ key-value pairs}$

Algorithm 2: QuickSelect (continued)(A, i)

1 if $i < n_{<}$ then _____
2 else if $i \geq n_{<} + n_{=}$ then _____
3 else _____

Q: If rank $i < n_{<}$, what do we do?

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1 if $i < n_{<}$ then **return** QuickSelect($A_{<}, i$)
2 else if $i \geq n_{<} + n_{=}$ then **return** QuickSelect($A_{>}, i - n_{<} - n_{=}$)
3 else _____

Q: Otherwise?

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Algorithm 2: QuickSelect (continued)(A, i)

```
1 if  $i < n_{<}$  then return QuickSelect( $A_{<}, i$ )
2 else if  $i \geq n_{<} + n_{=}$  then return QuickSelect( $A_{>}, i - n_{<} - n_{=}$ )
3 else return  $A_{=}[0]$ 
```

Any key-value pair in $A_{=}$ is valid output.

Example: the first pivot

Keys (ignore values):

$$A = (20, 30, 50, 60, 30, 50, 40), \quad i = 3.$$

Q: We choose index $p = 2$, so $\text{pivot} = 50$. What are the sub-arrays?

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$$A_{<} = (20, 30, 30, 40), \quad A_{=} = (50, 50), \quad A_{>} = (60).$$

Q: What call do we make next?

$$i = 3 < n_{<} = 4 \implies \text{QuickSelect}((20, 30, 30, 40), 3).$$

We still want rank 3, the fourth-smallest key in this smaller array.

Example: the next pivot

Consider `QuickSelect((20, 30, 30, 40), 3)`. Suppose its random pivot key is now `pivot = 30`.]

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$$A_{<} = (20), \quad A_{=} = (30, 30), \quad A_{>} = (40).$$

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$\text{QuickSelect}((40), 0)$ returns 40.

QuickSelect: correctness

We prove correctness for *every* sequence of pivot choices, using strong induction on the array size.

$P(k)$: For every array of size k and every valid rank i ,
QuickSelect(A, i) returns a valid key-value pair of rank i .

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- Note: each recursive call considers a smaller sub-array.

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- $n_{<} + n_{=} \leq i < n$: skip the first $n_{<} + n_{=}$ positions. The desired rank inside $A_{>}$ is $i - n_{<} - n_{=}$. By induction, the recursive call is correct.

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Q: What happens if every key equals the pivot?

We return from $A_{=}$ after one partitioning pass.

QuickSelect: various runs

Suppose the keys are distinct and we ask for the maximum, $i = n - 1$.

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$$n, n - 1, n - 2, \dots, 1.$$

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Each partitioning pass takes time proportional to the current array size.

$$n + (n - 1) + \dots + 1 = \Theta(n^2).$$

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Q: But what if the pivot is the largest key?

One iteration suffices with $O(n)$ runtime.

The running time depends on the random choices.

QuickSelect: expected shrinkage

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Q: How large is this bound on average over a uniformly random pivot?

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$$M \leq \max\{n_<, n_>\}.$$

Q: How large is this bound on average over a uniformly random pivot?

$$\mathbb{E}[M] \leq \frac{3n}{4}.$$

(We will not do this calculation; all you need to know is that expected size drops).

Intuitively, this means that the expected runtime satisfies $T_{\text{expected}}(n) \leq cn + T_{\text{expected}}(3n/4)$.

By unrolling, this can be solved to give $T_{\text{expected}}(n) \leq 4cn$.

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Q: Why is QuickSelect a Las Vegas algorithm?

Every pivot sequence gives a correct answer, and the expected running time is $O(n)$.

An individual run may still take $\Theta(n^2)$ time.

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$$T_{\text{expected}}(n) = \max_{x: \text{size}(x) \leq n} \mathbb{E}[T_A(x)].$$

For a nonnegative integer-valued random variable Z ,

$$\mathbb{E}[Z] = \sum_{z \in \mathbb{N}} z \cdot \Pr[Z = z].$$

Fix an input, average over the random choices, then take the worst input.

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- If $f(x) \neq \emptyset$, then

$$\Pr[A(x) \in f(x)] \geq 1 - p.$$

Monte Carlo algorithms

A *Monte Carlo algorithm* always runs within a chosen time bound, but may give an incorrect answer with small probability. For a computational problem $\Pi = (\mathcal{I}, \mathcal{O}, f)$, error probability at most p means that, for every input $x \in \mathcal{I}$,

- If $f(x) \neq \emptyset$, then

$$\Pr[A(x) \in f(x)] \geq 1 - p.$$

- If $f(x) = \emptyset$, then

$$\Pr[A(x) = \perp] \geq 1 - p.$$

Las Vegas and Monte Carlo

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QuickSelect is Las Vegas. Next: a Monte Carlo sampling algorithm.

EstimateProportion

A pollster wants to estimate the fraction of people who prefer singing to dancing. **Input:**

$A = (t_0, \dots, t_{n-1})$, where $n \geq 1$, $t_j \in \{S, D\}$, and an error tolerance $\varepsilon \in (0, 1)$.

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Output:

A rational number $r \in [0, 1]$ such that

$$|r - \theta| \leq \varepsilon.$$

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Algorithm 3: EstimateBySampling(A, ε)

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1 Set  $m = \lceil 25/\varepsilon^2 \rceil$  and  $s = 0$ 
2 foreach  $t = 1, \dots, m$  do
3   | Draw  $j = \text{random}(n)$  independently
4   | if  $A[j] = S$  then Set  $s = s + 1$ 
5 return  $r = s/m$ 
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Sampling is *with replacement*: the same index may be chosen again. Assume a random index and access to $A[j]$ each take $O(1)$ time.

The sampling guarantee

With $m = \lceil 25/\varepsilon^2 \rceil$ independent samples, a probability calculation gives

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- This holds for every fixed array A .
- The algorithm always takes $O(1/\varepsilon^2)$ time in our access model.
- It is a Monte Carlo algorithm: some random samples give a bad estimate.

Power of randomness

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Conjecture:

randomness is not necessary for polynomial-time computation.